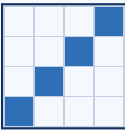


Three regimes, set by the modulus

The argument θ fixes how fast the plane turns and never affects whether it grows. That is decided entirely by the modulus, and the three cases below are the only ones there are.

NEITHER GROWING NOR DECAYING

1



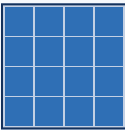
On the unit circle | $r = 1$

GEOMETRIC ACTION	DET OF THE 2×2 BLOCK	ORBIT OF A POINT
pure rotation by θ , no scaling	$\det A = 1$	a closed circle

Lengths are preserved exactly, so the matrix is **orthogonal**. Every orbit closes if θ is a rational multiple of 2π and fills the circle densely otherwise — the boundary case in every stability argument.

GROWING OR DECAYING

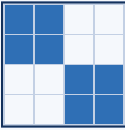
2



Outside the unit circle | $r > 1$

GEOMETRIC ACTION	DET OF THE 2×2 BLOCK	ORBIT OF A POINT
rotation with outward scaling by r	$\det A > 1$	a spiral outward

Each application turns by θ and stretches by r , so iterates diverge while circling. In a discrete system this is instability — and the divergence is geometric, so it is fast.



Inside the unit circle | $r < 1$

GEOMETRIC ACTION	DET OF THE 2×2 BLOCK	ORBIT OF A POINT
rotation with inward scaling by r	$\det A < 1$	a spiral inward to the origin

The stable case for iteration: every orbit converges to the origin, still turning as it goes. This is what $|\lambda| < 1$ means geometrically, and why the condition is stated on the modulus rather than on the real part.

The determinant of the 2×2 block is $a + b = r$, so the **area scaling** and the modulus are the same fact stated twice — which is why the determinant can be read as a stability marker without computing an eigenvalue at all.